OVERVIEW OF VOICE, VIDEO, and DATA COMMUNICATION

Topics to be covered

I. Introduction to data communication
II. Four fundamental characteristics a data communications system
III. Five components of a data communications system
IV. Different forms of data representation (Text, Numbers, Images, Audio, & Video)
V. Different modes of data flow

I. Introduction to Data Communication

- **Communication** : means sharing information
  - Locally
  - Remotely (over distance)

Differences between Communication and Transmission

Communication	Transmission
• It refers to exchange of meaningful information between two communicating devices.	• It refers to the physical movement of information.
• It is a two-way scheme.	• It is a one-way scheme.

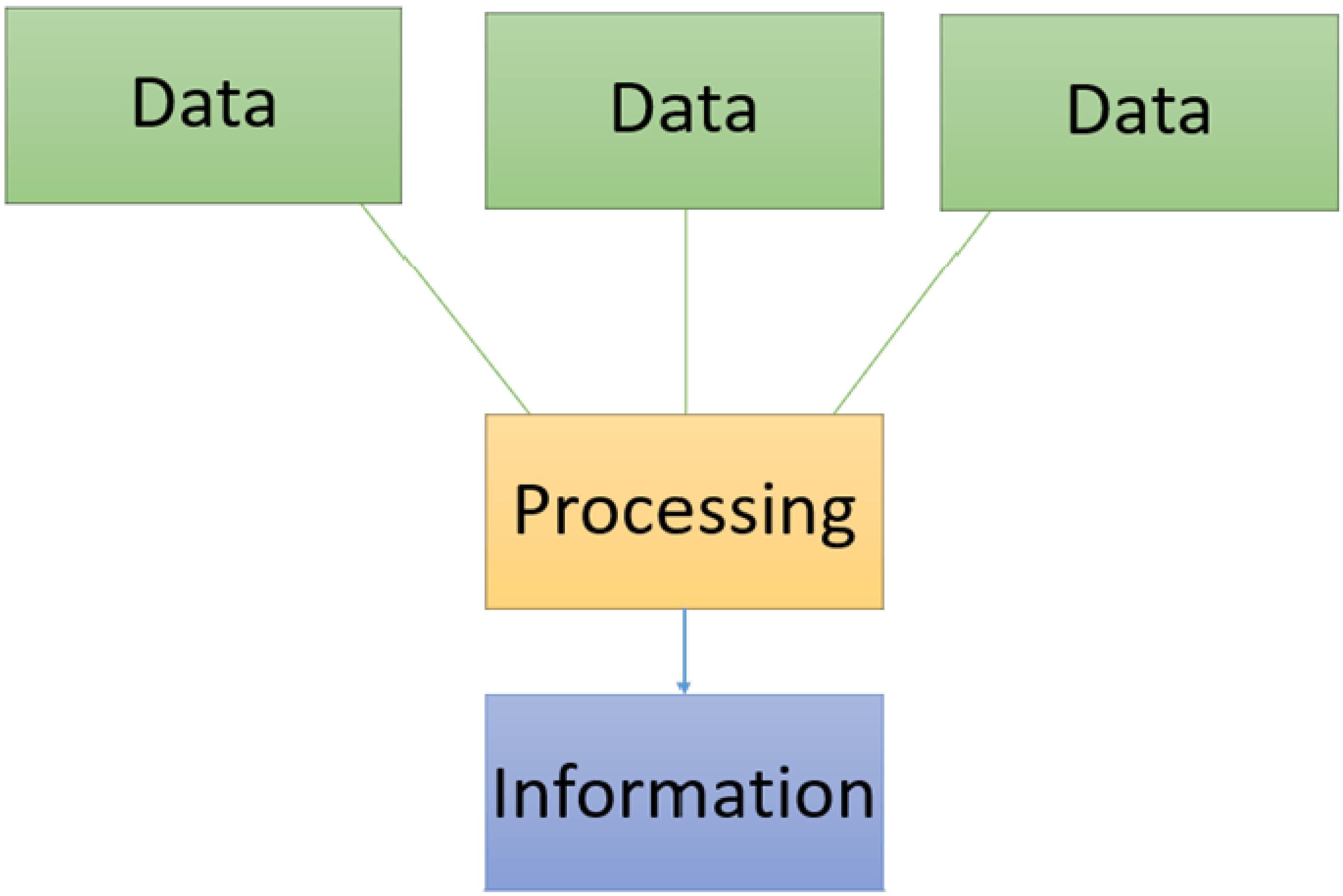
I. Introduction to Data Communication

- **Data** : Refers to information presented in whatever form is agreed upon by the parties creating and using data.

I. Introduction to Data Communication

Data	Information
Data is unorganized raw facts that need processing to make it meaningful	Information is a processed, organized data presented in a given context to make it meaningful and useful
Data comprises facts, observations, perceptions, numbers, characters, symbols, image, etc	Information is processed data that possess context, relevance, and purpose
Data never depends on information	Information is dependent on data
Data is measured in bits and bytes	Information is measured in meaningful units like time, quantity, etc
Data can be structured, tabular data, graph, data tree	Information is language, ideas, and thoughts based on the given data

I. Introduction to Data Communication



I. Introduction to Data Communication

Data

- 100

Information

- 100 miles

Knowledge

- 100 miles is quite a far distance.

Wisdom

- It is very difficult to walk 100 miles by any person, but vehicle transport is okay

I. Introduction to Data Communication

- **Data communications**: the **exchange** of data between two devices via some form of wired or wireless transmission medium.
- **Communication system** : a combination of **hardware** (physical equipment) and **software** (programs)

II. Four fundamental characteristics of a data communications system

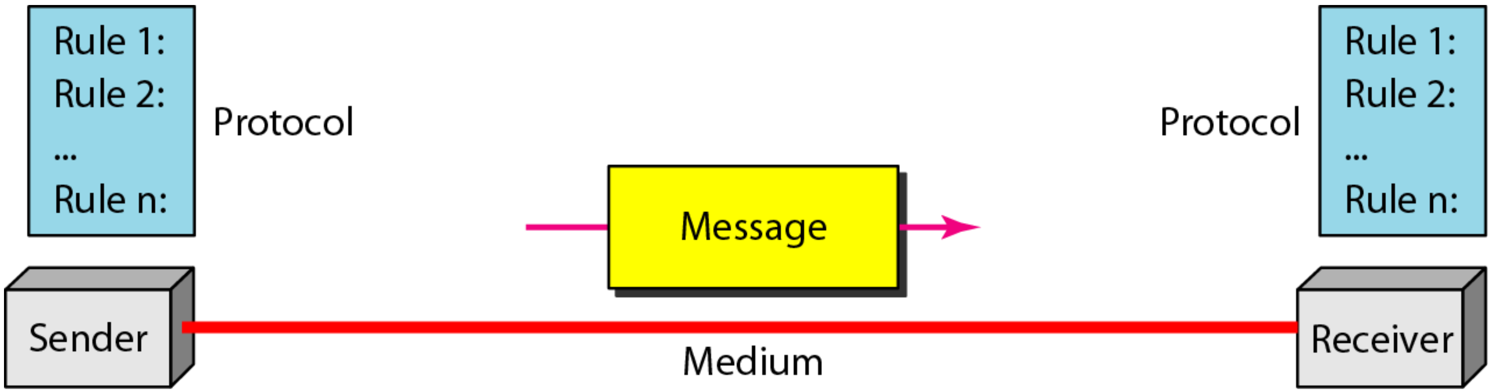
- The effectiveness of a data communications system depends on :
  1. Delivery
  2. Accuracy
  3. Timelines
  4. Jitter *(variation in packet arrival time)*

II. Four fundamental characteristics of a data communications system

- The effectiveness of a data communications system depends on :

1. **Delivery.** The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.
2. **Accuracy**. The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.
3. **Timeliness**. The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called *real-time transmission.*
4. **Jitter**. Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets. For **example**, let us assume video packets are sent every 30 ms. If some packets arrive with 30-ms delay, and others with 40-ms delay, an uneven quality in the video is the result.

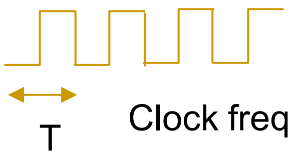
III. Five components of a data communications system



IV. Different forms of Data Representation

1. Text - ASCII constitutes the first 127 characters in Unicode
2. Numbers - binary number
3. Images
4. Audio
5. Video

Data Transfer Rate



Clock frequency (Hz)

$$T = \frac{1}{f}\ \text{sec}$$

$$T = \frac{1}{8\times 10^{6}} = 0.000000125\ \text{sec}$$
$$= 0.125\ \mu\text{s}$$

$$\text{Data transfer rate (bps)} = \frac{\text{Number of bits transmitted per operation (bits)}}{\text{Transfer time per operation (s)}}$$

$$\text{Data transfer rate} = 16\times 8\times 10^{6} = 128\times 10^{6}\ \text{bps} = 128\ \text{Mbps}$$

$$\text{Data transfer rate} = 128\ \text{Mbps}$$
$$= \frac{128}{8}\ \text{Mbps} = 16\ \text{MB/s}$$

DATA FORMATS: Numbers - Binary and Hex

	128	64	32	16	8	4	2	1	Decimal
	b_7	b_6	b_5	b_4	b_3	b_2	b_1	b_0	
Val1	0	0	1	1	0	1	0	1	53
Val2	0	1	1	0	1	0	1	0	106
Val3	0	0	0	1	1	0	1	0	26

What is 0010 1011 in decimal?

What is 93 in binary?

Val	Hex		Val	Hex
0000	0		1000	8
0001	1		1001	9
0010	2		1010	A
0011	3		1011	B
0100	4		1100	C
0101	5		1101	D
0110	6		1110	E
0111	7		1111	F

What is 0010 1011 in hex?

DATA FORMATS: Codes - ASCII

Standard ASCII Chart/Table

Dec	Hex	Name	Char	Ctrl-char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
0	0	Null	NUL	CTRL-@	32	20	Space	64	40	@	96	60	`
1	1	Start of heading	SOH	CTRL-A	33	21	!	65	41	A	97	61	a
2	2	Start of text	STX	CTRL-B	34	22	"	66	42	B	98	62	b
3	3	End of text	ETX	CTRL-C	35	23	#	67	43	C	99	63	c
4	4	End of xmit	EOT	CTRL-D	36	24	$	68	44	D	100	64	d
5	5	Enquiry	ENQ	CTRL-E	37	25	%	69	45	E	101	65	e
6	6	Acknowledge	ACK	CTRL-F	38	26	&	70	46	F	102	66	f
7	7	Bell	BEL	CTRL-G	39	27	'	71	47	G	103	67	g
8	8	Backspace	BS	CTRL-H	40	28	(	72	48	H	104	68	h
9	9	Horizontal tab	HT	CTRL-I	41	29	)	73	49	I	105	69	i
10	0A	Line feed	LF	CTRL-J	42	2A	*	74	4A	J	106	6A	j
11	0B	Vertical tab	VT	CTRL-K	43	2B	+	75	4B	K	107	6B	k
12	0C	Form feed	FF	CTRL-L	44	2C	,	76	4C	L	108	6C	l
13	0D	Carriage feed	CR	CTRL-M	45	2D	-	77	4D	M	109	6D	m
14	0E	Shift out	SO	CTRL-N	46	2E	.	78	4E	N	110	6E	n
15	0F	Shift in	SI	CTRL-O	47	2F	/	79	4F	O	111	6F	o
16	10	Data line escape	DLE	CTRL-P	48	30	0	80	50	P	112	70	p
17	11	Device control 1	DC1	CTRL-Q	49	31	1	81	51	Q	113	71	q
18	12	Device control 2	DC2	CTRL-R	50	32	2	82	52	R	114	72	r
19	13	Device control 3	DC3	CTRL-S	51	33	3	83	53	S	115	73	s
20	14	Device control 4	DC4	CTRL-T	52	34	4	84	54	T	116	74	t
21	15	Neg acknowledge	NAK	CTRL-U	53	35	5	85	55	U	117	75	u
22	16	Synchronous idle	SYN	CTRL-V	54	36	6	86	56	V	118	76	v
23	17	End of xmit block	ETB	CTRL-W	55	37	7	87	57	W	119	77	w
24	18	Cancel	CAN	CTRL-X	56	38	8	88	58	X	120	78	x
25	19	End of medium	EM	CTRL-Y	57	39	9	89	59	Y	121	79	y
26	1A	Substitute	SUB	CTRL-Z	58	3A	:	90	5A	Z	122	7A	z
27	1B	Escape	ESC	CTRL-[	59	3B	;	91	5B	[	123	7B	{
28	1C	File separator	FS	CTRL-\	60	3C	<	92	5C	\	124	7C	|
29	1D	Group separator	GS	CTRL-]	61	3D	=	93	5D	]	125	7D	}
30	1E	Record separator	RS	CTRL-^	62	3E	>	94	5E	^	126	7E	~
31	1F	Unit separator	US	CTRL-_	63	3F	?	95	5F	_	127	7F	DEL

Boolean Logic Operations

AND			OR			X-OR		
A	**B**	**Z**	**A**	**B**	**Z**	**A**	**B**	**Z**
0	0	0	0	0	0	0	0	0
0	1	0	0	1	1	0	1	1
1	0	0	1	0	1	1	0	1
1	1	1	1	1	1	1	1	0

Bit shifts

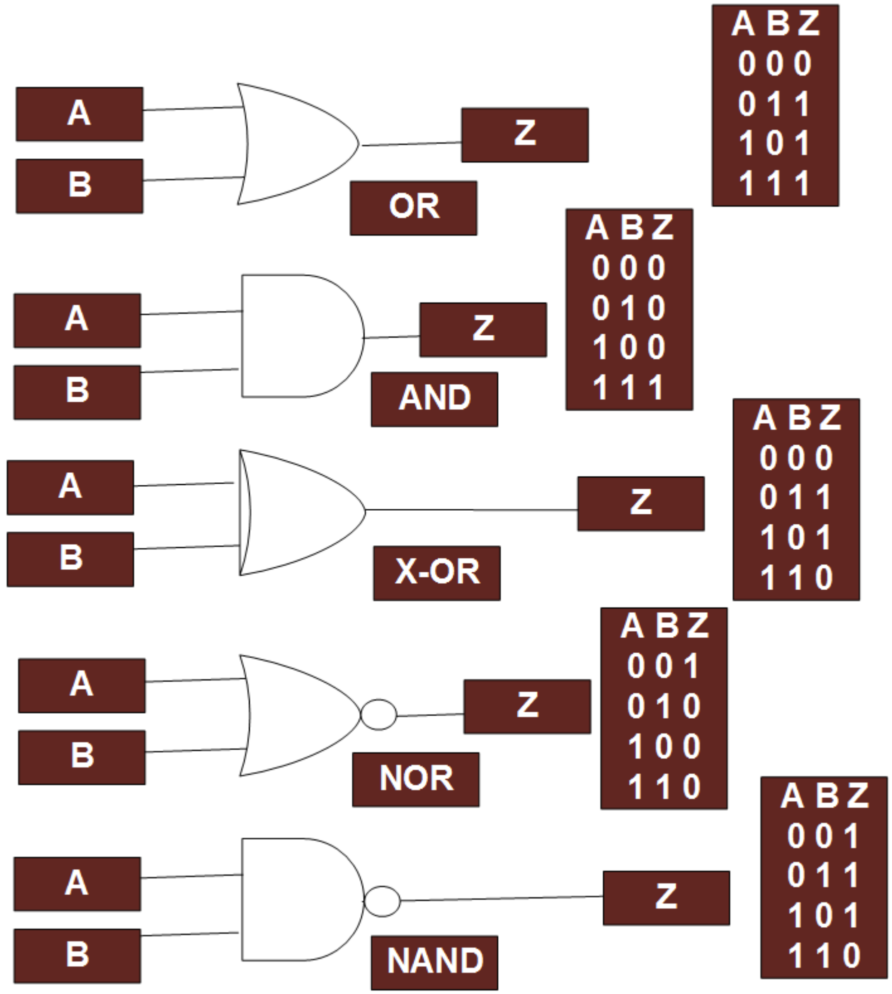
	128	64	32	16	8	4	2	1	Decimal
	b_7	b_6	b_5	b_4	b_3	b_2	b_1	b_0	
Val	0	0	1	1	0	1	0	1	53
Val << 1	0	1	1	0	1	0	1	0	106
Val >> 1	0	0	0	1	1	0	1	0	26

Bit masking

Value	0	0	1	1	0	1	0	1
AND	0	0	0	0	1	1	1	1
	----	----	----	----	----	----	----	----
	0	0	0	0	0	1	0	1

Result after Masking

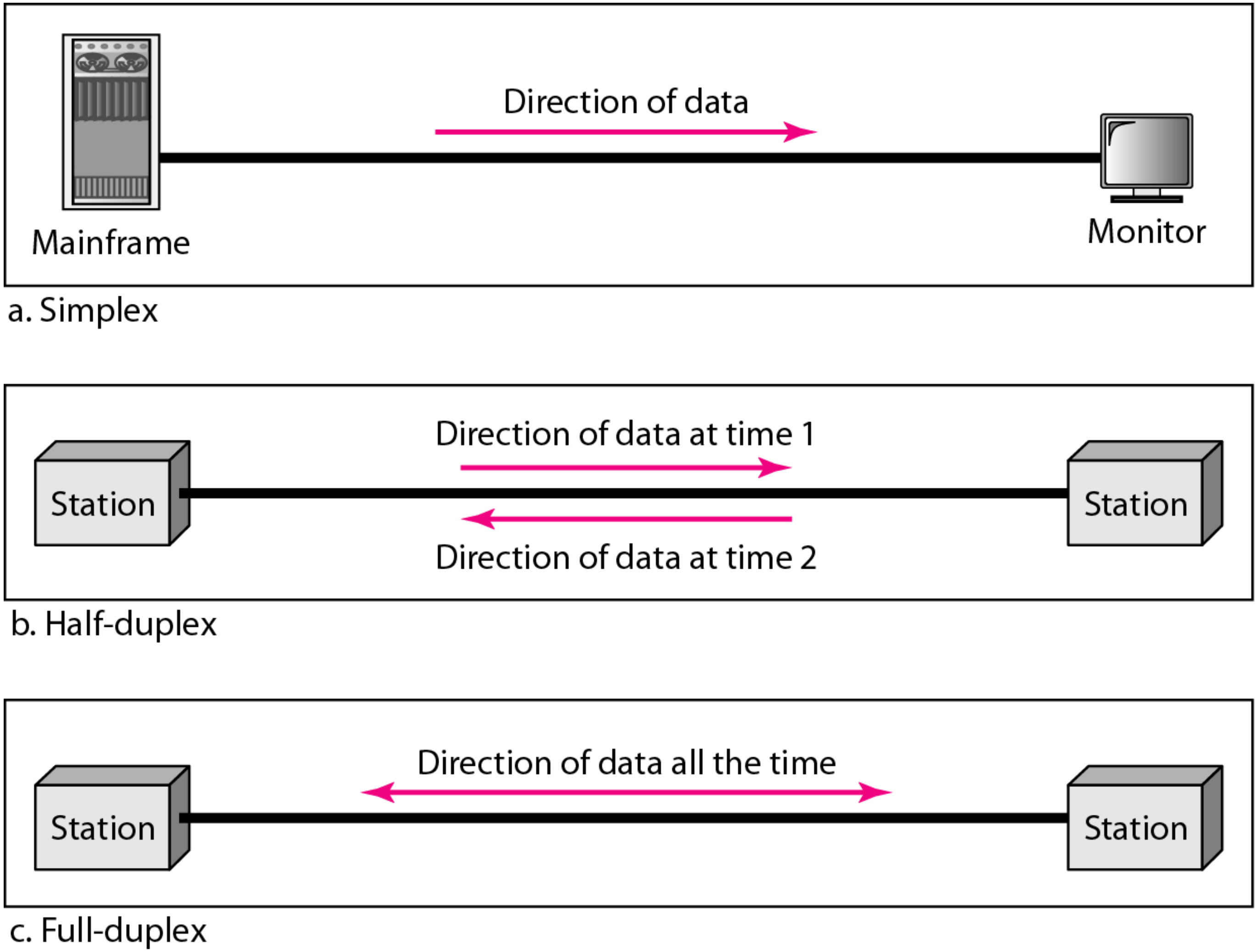
Logic gates


A
B
OR
Z
A B Z
0 0 0
0 1 1
1 0 1
1 1 1
A
B
AND
Z
A B Z
0 0 0
0 1 0
1 0 0
1 1 1
A
B
X-OR
Z
A B Z
0 0 0
0 1 1
1 0 1
1 1 0
A
B
NOR
Z
A B Z
0 0 1
0 1 0
1 0 0
1 1 0
A
B
NAND
Z
A B Z
0 0 1
0 1 1
1 0 1
1 1 0

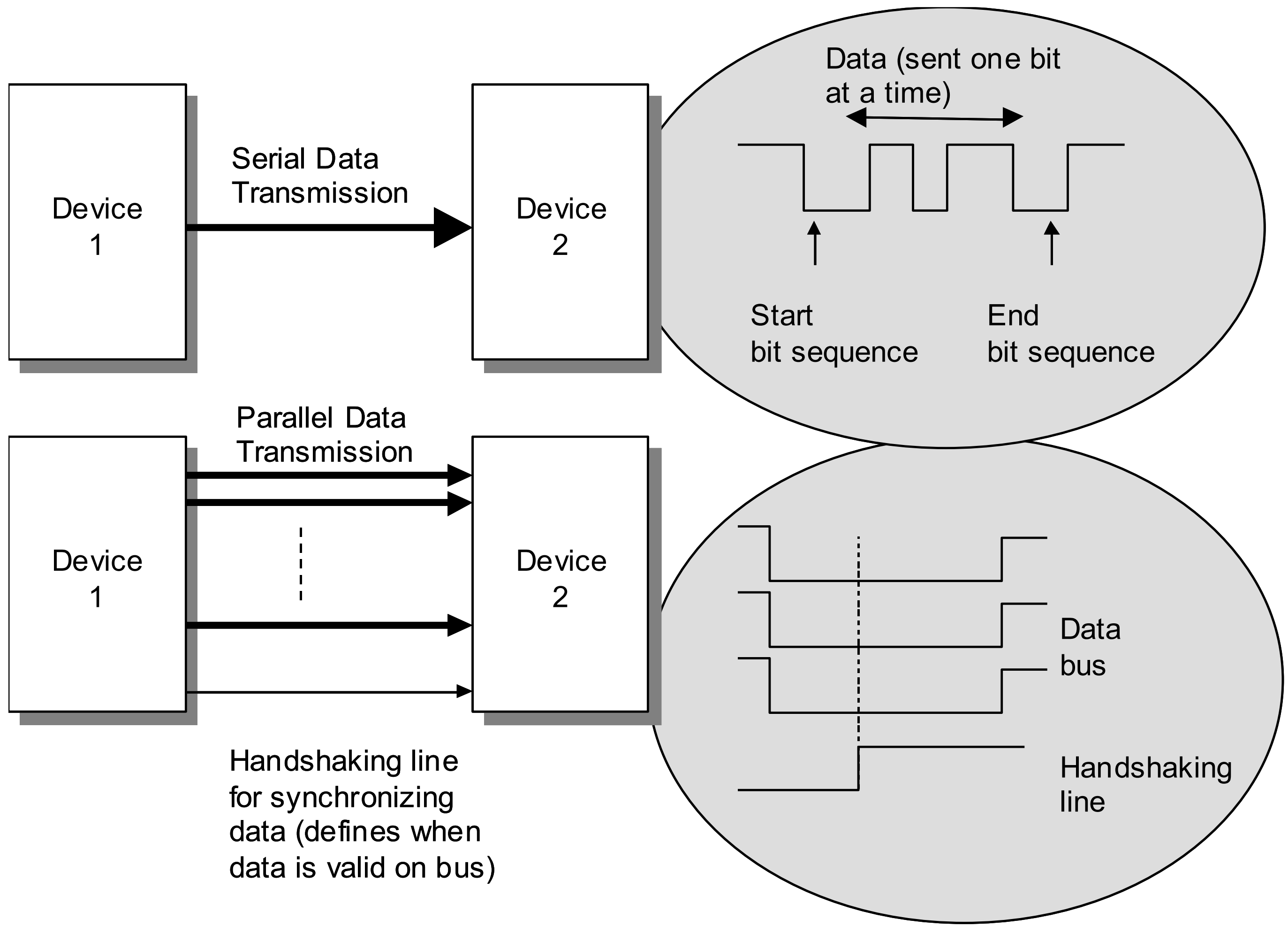
V. Different Modes of Data Flow

1. **Simplex** *:* The communication is unidirectional.
   - **E.g**., Keyboards(i/p) and traditional monitors(o/p) .
2. **Half-Duplex** : Each station can both transmit and receive, but not at the same time.
   - **E.g**., walkie-talkies and citizens band radios.
3. **Full-Duplex** (also called **duplex**) *:* Both stations can transmit and receive simultaneously.
   - **E.g.**, the telephone network.

Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



Serial or Parallel?



Thank You